

PORVOO HARABACKA, Finland

WMO: 027590

Lat: 60.3917N

Lon: 25.6072E

Elev: 77

StdP: 14.66

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-1.5	-11.7	2.9	-7.3	-5.9	4.0	-1.4	21.8	25.8	19.1	26.0	5.3	0	0.609

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	81.5	66.6	77.9	64.7	74.8	62.7	69.1	77.5	66.8	74.9	64.7	72.4	8.9	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.0	96.3	74.0	63.7	88.5	70.9	61.6	82.0	68.0	33.3	77.6	31.4	75.2	29.7	72.6	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.5	16.4	14.5	-13.3	85.1	7.0	3.9	-18.3	87.9	-22.4	90.2	-26.4	92.4	-31.4	95.2		
			-11.8	70.4	9.4	2.4	-18.5	72.1	-24.0	73.5	-29.3	74.8	-36.1	76.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	21.1	24.0	29.6	39.8	51.7	58.6	64.4	61.3	53.6	42.0	35.8	29.1
	DBStd	16.35	11.58	10.98	7.78	5.88	7.42	5.79	5.69	5.05	5.62	6.38	7.53	9.62
	HDD50	3997	897	727	632	310	65	6	0	1	30	255	426	648
	HDD65	8246	1362	1147	1097	756	414	205	83	137	343	713	876	1113
	CDD50	1326	0	0	0	4	120	264	445	349	137	7	0	0
	CDD65	100	0	0	0	0	3	13	63	21	0	0	0	0
	CDH74	842	0	0	0	0	49	128	521	144	0	0	0	0
	CDH80	144	0	0	0	0	4	14	103	23	0	0	0	0
Wind	WSAvg	7.4	7.6	7.6	7.8	7.2	7.4	7.7	6.9	6.7	6.7	7.5	7.7	8.3
Precipitation	PrecAvg	26.7	2.3	1.7	1.4	1.3	1.6	2.6	2.5	2.9	2.5	2.8	3.0	2.3
	PrecMax	34.7	4.6	4.3	2.6	2.8	2.9	5.3	6.3	5.9	6.3	6.9	7.7	5.8
	PrecMin	21.7	0.4	0.1	0.2	0.2	0.2	0.7	0.0	0.2	0.5	0.7	0.5	0.6
	PrecStd	3.7	1.0	1.0	0.6	0.7	0.8	1.3	1.5	1.4	1.5	1.4	1.4	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	43.0	50.2	66.3	79.4	82.2	86.7	83.3	70.4	57.7	51.0	47.1
		MCWB	39.3	37.7	42.0	51.0	60.8	65.2	67.7	67.2	61.5	52.5	49.0	45.4
	2%	DB	38.8	38.8	45.3	60.8	74.3	77.6	82.7	77.8	67.2	55.0	48.3	43.8
		MCWB	37.9	36.6	38.8	48.5	57.9	62.6	67.6	65.7	60.0	51.7	47.3	42.1
	5%	DB	37.0	36.9	41.7	55.5	70.7	74.3	79.7	74.1	64.9	53.2	46.4	41.6
		MCWB	36.3	35.6	37.2	45.3	56.1	60.4	66.5	63.4	58.4	51.0	45.2	40.3
	10%	DB	34.9	35.5	38.5	51.1	66.5	70.8	76.5	71.2	62.7	51.5	44.9	39.7
		MCWB	34.1	34.5	35.5	42.7	54.1	59.2	65.1	61.7	57.7	49.4	43.8	38.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.6	38.5	42.9	51.9	64.0	67.8	71.6	70.9	63.1	54.8	49.3	46.2
		MCDB	40.1	40.5	47.9	62.6	75.1	76.7	80.9	79.4	67.9	56.6	50.9	47.0
	2%	WB	38.0	37.1	39.7	49.2	60.5	65.0	70.1	67.4	61.5	53.0	47.3	42.4
		MCDB	38.6	38.5	43.8	58.7	68.8	74.1	78.7	73.6	65.2	54.1	48.2	43.3
	5%	WB	36.4	36.0	37.8	46.6	58.0	62.4	68.4	65.1	60.2	51.5	45.4	40.5
		MCDB	37.0	37.1	40.4	53.4	67.2	71.4	76.9	71.6	63.4	52.6	46.1	41.4
	10%	WB	34.3	34.6	36.1	44.1	55.3	60.3	66.2	63.4	58.6	49.7	44.0	38.6
		MCDB	34.9	35.5	38.0	49.8	64.9	68.4	74.6	69.2	61.7	51.1	44.8	39.6
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.2	12.1	15.7	18.3	16.0	16.2	15.4	13.3	10.3	7.2	7.6
		MCDBR	6.0	7.5	15.2	25.3	24.7	21.8	20.7	20.5	16.8	12.0	7.6	7.7
		MCWBR	5.8	6.3	10.1	13.9	11.3	9.7	8.4	10.2	10.4	9.4	7.2	7.3
	5% WB	MCDBR	6.2	7.0	13.0	21.0	21.3	19.5	18.3	16.8	14.2	9.5	7.4	7.1
		MCWBR	6.1	6.1	9.4	12.5	11.0	10.0	8.4	9.2	9.5	8.7	7.2	6.9
Clear-Sky Solar Irradiance	taub	0.276	0.310	0.334	0.364	0.362	0.359	0.380	0.372	0.346	0.332	0.303	0.383	
	taud	2.133	2.182	2.243	2.334	2.406	2.455	2.414	2.434	2.498	2.391	2.209	2.604	
	Ebn at Noon	148	212	249	264	274	277	266	259	247	207	137	103	
	Edh at Noon	13	23	30	33	33	32	33	30	23	19	13	9	
All-Sky Solar Radiation	RadAvg	78	260	732	1218	1651	1715	1665	1283	780	339	95	45	
	RadStd	14	54	94	126	178	135	153	134	101	55	17	7	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (61 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	-256	N/A	N/A	N/A

Nomenclature: See separate page